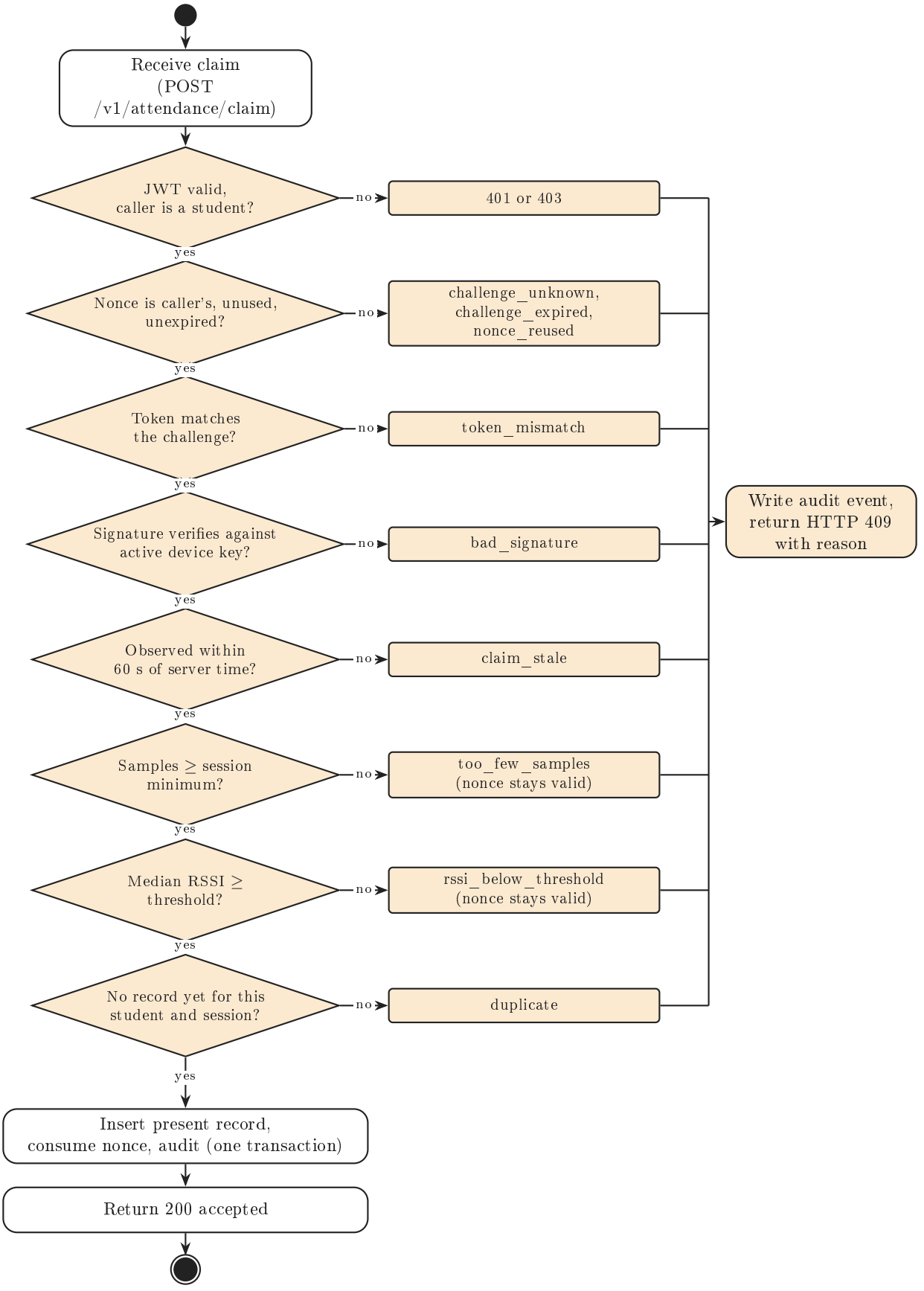
Receive claim (POST /v1/attendance/claim)
JWT valid, caller is a student?
no
401 or 403
yes
Nonce is caller's, unused, unexpired?
no
challenge_unknown, challenge_expired, nonce_reused
yes
Token matches the challenge?
no
token_mismatch
yes
Signature verifies against active device key?
no
bad_signature
yes
Observed within 60 s of server time?
no
claim_stale
yes
Samples ≥ session minimum?
no
too_few_samples (nonce stays valid)
yes
Median RSSI ≥ threshold?
no
rssi_below_threshold (nonce stays valid)
yes
No record yet for this student and session?
no
duplicate
yes
Write audit event, return HTTP 409 with reason
Insert present record, consume nonce, audit (one transaction)
Return 200 accepted